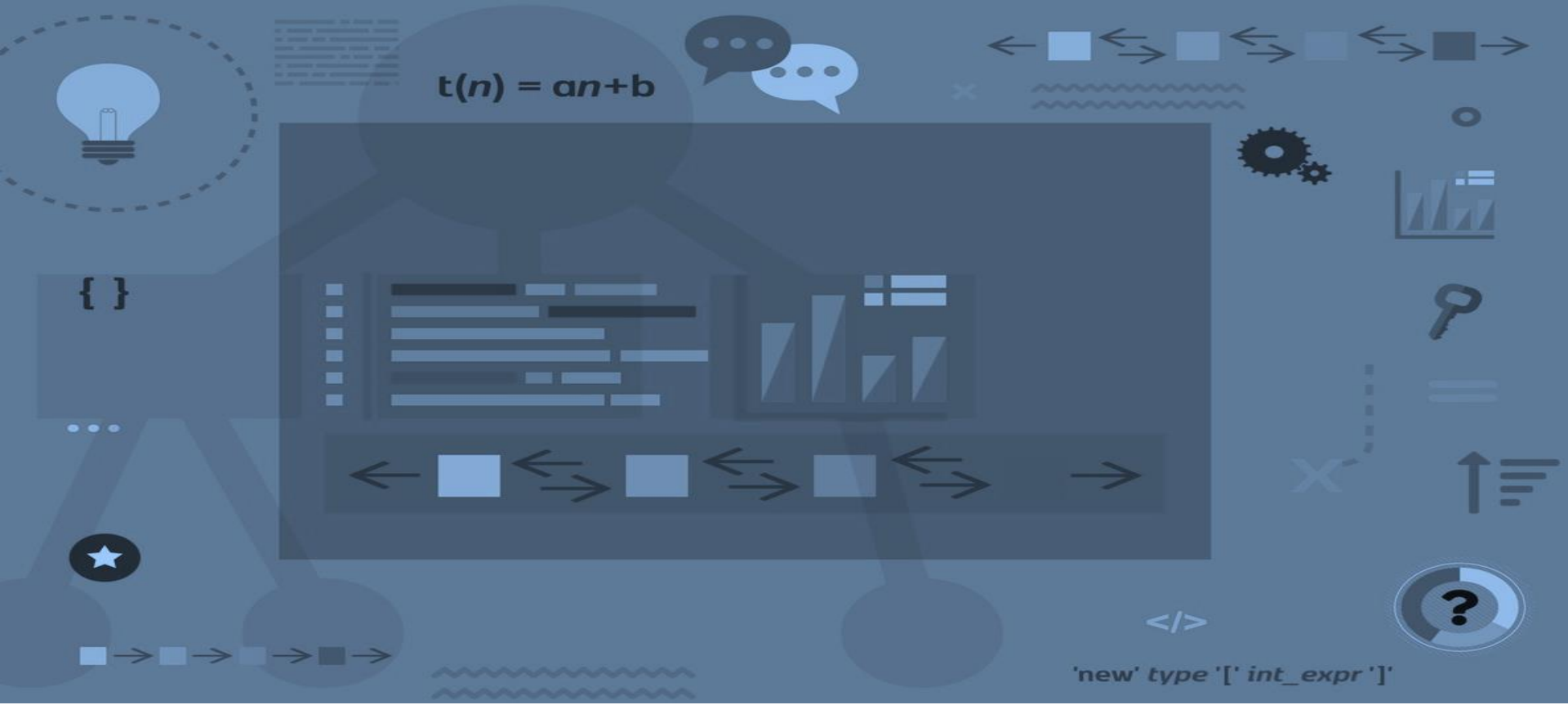

Java Programming

CPT111 – Lecture 0

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西交利物浦大学

CPT111 Java Programming

Lecture 0

Introduction & Course Logistics

Hi CPT111 Students !

- Welcome to CPT111!
  - In this course, you will learn how to program in Java and to do problem solving
- Academic calendar:

Semester	Month	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Description
		28	29	30	31	1	2	3	
Semester 1	September	4	5	6	7	8	9	10	Week 0: Registration and induction
		11	12	13	14	15	16	17	Week 1: Teaching starts
		18	19	20	21	22	23	24	Week 2
	October	25	26	27	28	29	30	1	Week 3; Mid-autumn day; National day
		2	3	4	5	6	7	8	National day; University closure days
		9	10	11	12	13	14	15	Week 4
		16	17	18	19	20	21	22	Week 5
		23	24	25	26	27	28	29	Week 6
	November	30	31	1	2	3	4	5	Week 7
		6	7	8	9	10	11	12	Week 8
		13	14	15	16	17	18	19	Week 9
		20	21	22	23	24	25	26	Week 10
	December	27	28	29	30	1	2	3	Week 11
		4	5	6	7	8	9	10	Week 12
		11	12	13	14	15	16	17	Week 13
		18	19	20	21	22	23	24	Reading Week
		25	26	27	28	29	30	31	Christmas; Reading Week
	January	1	2	3	4	5	6	7	New Year; Examination days
		8	9	10	11	12	13	14	Examination days

In-Class Quiz

- How are you doing today?
  a) I am well and ready to continue studying!
  b) I am sleepy and still in holiday mode
  c) What are we doing, AI will replace us anyway...

Course Goals and Prerequisites

- We're going to learn how to:
  - Write object-oriented code
    - Solve computation problems
    - Use IDE: NetBeans
  - Test and debug your code
    - Write correct code
    - Satisfy test cases
  - Design and document programs
    - Satisfy specification
    - Good documentation
- No prerequisites

Teaching Team

- Module Leader: Dr Erick Purwanto
  - erick.purwanto@xjtlu.edu.cn SD551 After lectures
- Teacher:
  - Dr Teng Ma teng.ma@xjtlu.edu.cn SD459 Friday 13-15

- Teaching Assistants :
  - Yida, Wenya, Wenlei, Xinyue, Luyan, Siqi,

Zihan, Ming, Ifan, Leonel, Ningfei, Haoyue

Timetable

	Monday	Tuesday	Wednesday	Thursday		Friday
9 AM						
10 AM						
11 AM						
12 PM						
1 PM				CPT111-Comp.Lab-D1/4 Erick Purwanto, TENG MA SIP-SD554 Week: 1-3,4-13	CPT111-Comp.Lab-D1/1 Erick Purwanto, TENG MA SIP-SD546 Week: 1-3,4-13	
2 PM						
3 PM				CPT111-Comp.Lab-D1/5 Erick Purwanto, TENG MA SIP-SD554 Week: 1-3,4-13	CPT111-Comp.Lab-D1/2 Erick Purwanto, TENG MA SIP-SD546 Week: 1-3,4-13	
4 PM	CPT111-Lecture-D1/1 Erick Purwanto, TENG MA SIP-SB123 Week: 1-3,4-13					
5 PM		CPT111-Lecture-D1/2 Erick Purwanto, TENG MA SIP-SD102 Week: 1-3,4-13		CPT111-Comp.Lab-D1/6 Erick Purwanto, TENG MA SIP-SD554 Week: 1-3,4-13	CPT111-Comp.Lab-D1/3 Erick Purwanto, TENG MA SIP-SD546, SIP-SD446E Week: 1-3,4-13	
6 PM						
7 PM						
8 PM						

Schedule

- On-site
  - Lectures: Monday 16:00 – 18:00 SB123 (Erick's office hour after that)
    Monday 17:30 – 19:30 SD102 (Erick's office hour after that)
  - Labs: Thursday 13:00 – 15:00 SD554, SD546
    Thursday 15:00 – 17:00 SD554, SD546
    Thursday 17:00 – 19:00 SD554, SD546, SD446E

- Lecture attendance are **compulsory**, affect resit exam
  - Lecture **recording** will be released on Week 6 and Week 13
- Lab attendance and lab activities are **graded**
- Syllabus & Teaching Plan : See **Module Handbook**

Where to Get Information ?

- Learning Mall
  - Announcement
  - Lecture and Lab materials
  - Discussion Forum
  - In-class Quizzes
  - Lab Exercises
  - Coding Assignments

- Office Hour and Email
  - only for private matters

- Optional Textbooks
  - Head First Java, 2nd Edition, by Kathy Sierra, Bert Bates
  - Effective Java, 3rd Edition, by Joshua Bloch

Assessment

1. **Coursework 1** : 15%
   - Weekly Coding Assignments
   - automatic graded in Learning Mall, solutions released after closing
2. **Coursework 2** : 15%
   - Lab Attendance and Lab Activities
3. **Coursework 3** : 30%
   - Individual Programming Project
   - Released Week 10, due Week 14, details TBA
4. **Final Exam** : 40%
   - Open Book, in Exam Week, details TBA

How do you learn?

- Learn **some** from
  - Listening to lectures
  - Reading the materials
- Learn **more** from
  - Trying out the codes from lectures and readings yourself
    - Discussing them with your friends / in forum
  - Re-doing the lecture codes and lab tasks
  - Doing the coding exercises
  - Experiment on your own
- You are encouraged to discuss the exercises with your study group
  - But write the code *yourself* !

Lab Attendance and Lab Activities

- **15% of your final grade**
- Hybrid Escape Room: online and onsite group activities
  - One group = 4 - 6 students
- Lab 1-6 and Lab 8-12:
  - Discuss tasks with your group members and TAs
  - Answer theory questions
  - Answer programming questions
  - Complete additional tasks
  - Receive a reward used in Lab 7 or Lab 13
- **Special activities in Lab 7 and Lab 13**
  - *Rescue the Captain Bird* and *Escape the Haunted Mansion*

CW1 - Coding Assignments

- **Graded,** part of your final grade : 15%
- Closed in 2 weeks, but ***start early!***
- Weekly coding assignments:
  - Attempts allowed: 1
  - Must pass **all** test cases
  - First check: no penalty
  - Second check and so on: cumulative 10% penalty
- **Solution is available after the closing time**
  - click Review on your submission
- Ask about the question if it is not clear in Discussion Forum
  - but **do not post** the solution code

Attending Lectures and Labs

- You will see explanation and sample codes in the slides
  - **Various tools** in Learning Mall
    - In-class Quiz
    - Autograded Quiz for Coding Exercises and CW1
  - Copy paste the sample codes into NetBeans in your computer
    - Modify and run it, **experiment with the code yourself** !

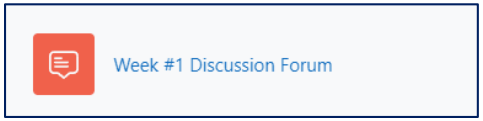
- **Complete** the slides
  - Print and scribble, or use a pdf annotation program
  - Recordings provided

- If you have further questions, post your questions in the Discussion Forum

Challenges and Forum

- Challenges :
  - It's an introduction coding course with non-trivial learning goals
  - We have a large cohort of ∼**500 students**

- Don't email, post your questions in the Discussion Forum



- You are also **encouraged to answer** your friends' questions
  - let us use forum to effectively answer your questions!

- Remember, **do not post** solution code
  - it will ruin other students learning development

Discussion Forum and General Attitude

- Write informative title

which one has helpful title?

Discussion	Started by	Last post	Replies
Q6 AnyRep	Tianxi Wen 5 Jun 2021	Erick Purwanto 6 Jun 2021	3
A little confusion about Exercise 11.2 ARSet EQUALS	Jingtian Zhang 27 May 2021	Erick Purwanto 27 May 2021	3
Why the for loop of items could not pass the test in 11.2 ecercise.	Junyi Wei 24 May 2021	Erick Purwanto 24 May 2021	1
Question4, 5, 7, 11, 14	Qinxin Ren 21 May 2021	Erick Purwanto 23 May 2021	1
Question about lec quiz 5, 6, 7, 8, 9	Ziling Dong 22 May 2021	Erick Purwanto 23 May 2021	1

Discussion Forum and General Attitude

- We are all still in the unpredicted and difficult pandemic situation
- Please in posting in the forum or email, always remember to:
  - **Be kind**
  - **Be patient**
  - **Be helpful**
- In addition:
  - **Search first**, maybe your friend has asked the same thing before
  - **Post in English**, be specific, provide enough information
    - For example, don't just write that some error happens, but explain what did you do that might cause it, write the error message, OS/browser you use
    - Provide *screenshot* if necessary

In-Class Quiz

- You can do the following in Discussion Forum, **except**:
  a) ask questions to clarify CW1 question
  b) request futher explanation on some lecture topic
  c) search and read other classmates' posts to be fully informed
  d) post your solution code

In-Class Quiz

- Are you planning to continue graduate study abroad?

  a) **yes**, I want to pursue Master's/PhD degree abroad

  b) **no**, I want to do something else

Your Long-term Goals

- Not just good grade
- Job interview, graduate study, recommendation letter

Recommendation Rankings

Please rank the applicant in all categories listed below.

I am comparing this applicant to:

Knowledge in subject of proposed study

Intellectual ability/Problem solving

Ability to learn

Ability to work with others

Responsiveness to feedback

Originality/Creativity

Logic/Analytical/Mathematical ability

English written communication

English oral communication

Perseverance towards goals

Laboratory skills (for lab-related disciplines)

- Exceptional (Top 2%)
- Superior (Top 5%)
- Excellent (Top 15%)
- Above Average (Top 25%)
- Average (Top 50%)
- Below Average (Lower 50%)
- Unable to Rate

Overall Recommendation

Recommendation Letter

In addition to the ratings above, University Graduate School requests a candid statement from you concerning this applicant. Please type the statement on official letterhead, if possible.

- Please explain in detail the applicant's accomplishments, abilities, character, and capacity for success as a graduate student in the proposed field of study.
- If the applicant's native language is not English, please comment on the applicant's English proficiency.

You may attach your recommendation letter below as a Word or PDF document.

Recommendation Letter

Choose File No file chosen

Thank you for your attention !

- Next, please go to **Installation Guide to install Java and NetBeans**
  - ask in the forum or during the **Lab 1 sessions** if you have questions
- And then, go to **Lecture 1** and start reading this week's material
- Finally, go to **Lab 1 sessions** to start exploring our tools and do this week's CW1s / online coding assignments...